Sexual Reproduction in Bacteria In bacterial sexual reproduction there is no meiosis, formation of gametes and zygote. Instead, it involves transfer of a portion of genetic material (DNA) from a donor cell to a recipient cell. This process is called as genetic recombination or parasexuality. It is known to occur in the following three ways:

Transformation

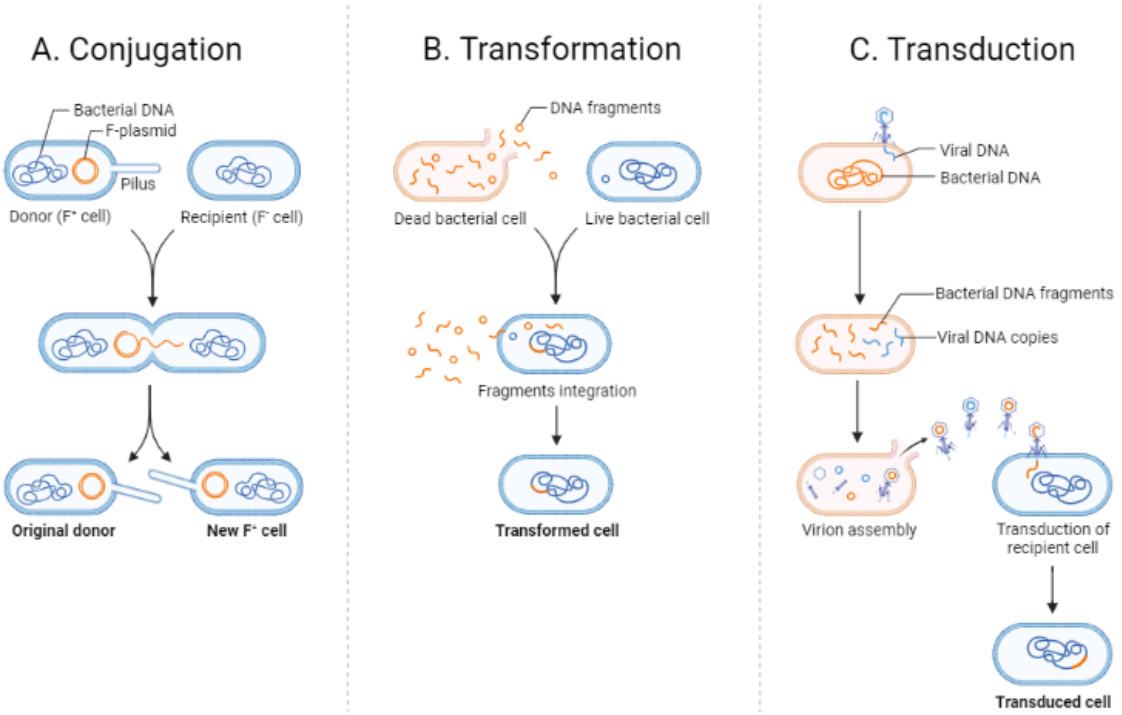
In this process, one kind of bacterium is transformed into another kind. It takes place by a transfer of DNA from a capsulated bacterium into a non-capsulated bacterium. It has been observed in *Diplococcus* bacteria.

Transduction

In this process, DNA of a bacterial cell (donor) is transferred into another bacterial cell with the help of a bacteriophage - *a type of virus that infects and replicates inside bacteria* This process is known to occur in several bacterial species such as *Salmonella, Escherichia, Micrococcus and Shigella.*

Conjugation

It is a process in which the genetic material of a bacterial cell of a particular strain is transferred into that of another bacterial cell of a different strain. Of the two strains of bacteria involved, one acts as a donor (or male) and the other as a recipient (or female). The donor cells are known to possess a sex factor or fertility factor (F factor) as a component of their circular DNA (F+ strain). The recipient cell does not have this factor, and hence it is described as F- F-strain. A conjugation between cells of F+ and F- strains always results in the formation of F+ bacterial cells in the progeny. *Offsprings or descendants produced by a parent organism or a particular cross of organisms.*


A. Conjugation
Bacterial DNA
F-plasmid
Pilus
Donor (F+ cell)
Recipient (F- cell)
Original donor
New F+ cell
B. Transformation
DNA fragments
Dead bacterial cell
Live bacterial cell
Fragments integration
Transformed cell
C. Transduction
Viral DNA
Bacterial DNA
Bacterial DNA fragments
Viral DNA copies
Virion assembly
Transduction of recipient cell
Transduced cell